

ASSIGNMENT #6

Course: CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2020
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Due Date: Saturday April 11th by 12:00 noon

Number of points.....100

Your score.....

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Scan your Assignment and create a PDF file
- PDF File Name: Assignment6_XY.PDF
 - **X:** First letter of your first name
 - **Y:** Your last name
 - Example: Assignment6_MCaracotsios.PDF
- Email your PDF file to: Suzane Carneiro svieir2@uic.edu by the due date (Saturday April 11) and time (12:00 noon)

Note: All problems are worth 100 points. Your final score will be the average value

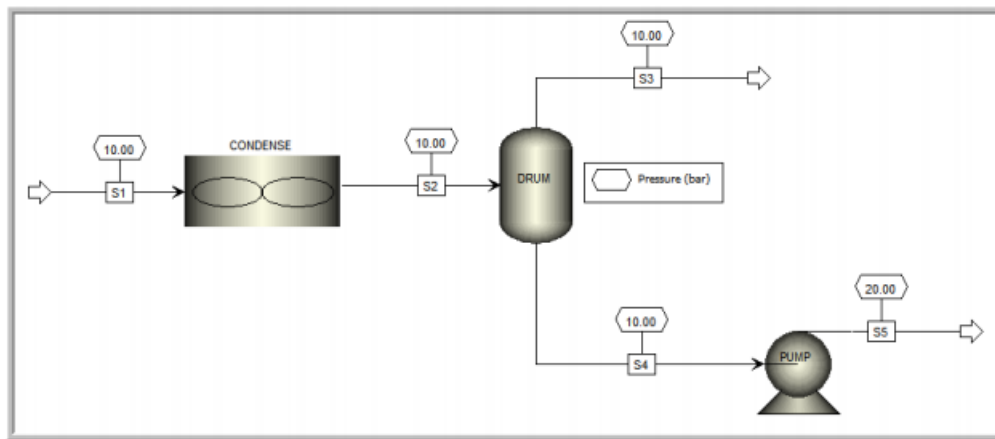
Problem 6A

A hot hydrocarbon feed stream consisting of propane(1), toluene(2), ethyl benzene(3) and p-xylene(4) enters a shell and tube condenser at temperature $T = 275^\circ\text{C}$ and pressure $P = 10\text{ bar}$ with an overall molar composition $[z_1, z_2, z_3, z_4] = [0.33, 0.30, 0.15, 0.22]$ and mass flow rate $F = 150,000 \frac{\text{kg}}{\text{hr}}$. The hydrocarbon mixture is then cooled under constant pressure until a liquid stream of volumetric flow rate $L = 700\text{ gpm}$ is produced. (See process flowsheet diagram below. The liquid is then pumped to an inventory tank for sales. The discharge pressure of the pump is set at $P = 20\text{ bar}$.

1. Calculate the bubble point and dew point temperature of the hydrocarbon feed mixture.
2. Calculate the temperature and molar composition of the liquid and vapor streams after the cooling process.
3. Calculate the amount of heat in kW removed from the feed stream during the cooling process.
4. Calculate the power requirements in kW for the pump if its efficiency is 75%.
5. Calculate the temperature rise across the pump.

For all calculations assume that both the vapor and liquid phases behave ideally and that Raoult's law applies for vapor liquid equilibrium calculations for all components except propane. Propane is at supercritical conditions so you must use Henry's law with coefficient:

$$H_1[\text{bar}] = \exp\left(9.13237 - \frac{1940.2}{T}\right) \quad T[=]\text{K}$$



Problem 6.A.1 Bubble Point and Dew Point Temperatures

Consider the hydrocarbon mixture described in the problem statement above, in the liquid phase. If we are to increase the temperature of the mixture, we reach a point at which we form a single bubble of vapor. This temperature is known as the bubble point temperature. Similarly if we consider the hydrocarbon mixture to be entirely vapor and then begin to lower the temperature until a single drop of liquid forms this temperature is known as the dew point temperature. In chapter 6 of the lecture notes alpha, α , was defined as the ratio of vapor fraction to the feed. It would follow that an alpha value of 1 would correspond to the dew point and a value of 0 would correspond to the bubble point. By utilizing Raoult's Law, the Rachford-Rice equation, and Henry's Law we can solve for these temperatures in the 301 Calculator.xlsm:

$$\sum_{i=1}^N \frac{z_i (K_i - 1)}{1 + \alpha (K_i - 1)} = 0 \quad \text{Rachford-Rice Equation}$$

$$K_i = \frac{P_i^{\text{Sat}}}{P} \quad \text{Raoult's Law}$$

$$K_i = \frac{H_i}{P} \quad \text{Henry's Law}$$

First we enter our 4 components into the tool then move to the VLE calculations tab. We then enter in the molar compositions, pressure and temperature. For components 2,3,4 we use Raoult's law to calculate the K-Values: $K_i = \frac{P_i^{\text{Sat}}}{P}$. Where the saturation pressure is given by the Psat function. Because $T = 275^\circ\text{C}$ is above the critical temperature of propane we have to rely on Henry's law for the K-Value of propane. Implementing the formula, $H_1[\text{bar}] = \exp\left(9.13237 - \frac{1940.2}{T}\right)$, $T[=]\text{K}$ into excel we then use Henry's law: $K_i = \frac{H_i}{P}$. At this point we set alpha to be 1 then using a computational method we can set the Rachford-Rice equation equal to zero by changing the temperature to find our dew point temperature. This same process is repeated for the bubble point temperature with an alpha value of 0, resulting in the temperatures:

$$T_{\text{Dew}} = 218.47 \text{ K}$$

$$T_{\text{Bubble}} = 65.38 \text{ K}$$

Fluid Conditions	Mixture	C3H8	C7H8	C8H10	C8H10
Molar Composition	1.0000	0.33	0.3	0.15	0.22
Temperature, C	218.4732062				
Pressure, bar	10.0000				
Saturated Vapor Pressure, bar		178.7255983	9.659630044	5.622310019	5.443243459
Saturated Vapor Temperature, C					
Molecular Weight	81.4765	44.0970	92.1410	106.1680	106.1680
K-Values		17.87255983	0.965963004	0.562231002	0.544324346
New K-Values					
Vapor to Feed Molar Ratio [a=V/F]	1.0000				
The Rachford-Rice Equation	0.0000				

Goal Seek ? X

Set cell: \$B\$14

To value: 0

By changing cell: \$B\$4

OK Cancel

Problem 6.A.2 Condenser Effluent Temperature

Passing through the condenser the mixture is cooled under constant pressure until the liquid stream reaches a volumetric flow rate of 700 gpm. This statement gives us two constraints, pressure and volumetric flow. A third constraint is that the Rachford-Rice Equation must be equal to zero at equilibrium. This leaves us with the ability to vary temperature and alpha in order to meet these constraints. The temperature under these constraints corresponds to the temperature of the condenser effluent. The following is the step-wise process followed using the 301 Calculator.xlsm:

If we begin where we left off with problem 6.A.1, we change our temperature to 275°C and guess an arbitrary alpha value (note $0 \leq \alpha \leq 1$). Now to setup the volumetric flow rate constraint we begin with the formula $1 - \alpha = \frac{L}{F}$. Solving for L, we implement this formula using the provided feed, $F = 150,000 \frac{\text{kg}}{\text{hr}}$. To convert this into gpm we have to setup a few conversion factors. First the average molecular weight can be calculated by taking the sum-product of the molecular weights of each component and the molar compositions. Next we set up the molar compositions in the liquid phase using the formula $x_i = \frac{Z_i}{1 + \alpha(K_i - 1)}$ and vapor phase using $y_i = K_i x_i$. We can find the volume of each component by taking the inverse of the Rackett function. This method works for all components except propane, which has to utilize the critical temperature in the data base. Taking the sum-product of the

liquid molar composition and molar volume we calculate the average molar volume of the mixture. Using these calculations along with some conversion factors we can convert our liquid flow to gpm:

$$\frac{x \text{ kg}}{\text{hr}} \frac{81.48 \text{ kmol}}{\text{kg}} \frac{0.1436 \text{ m}^3}{\text{kmol}} \frac{264.172 \text{ gal}}{\text{m}^3} \frac{1 \text{ hr}}{60 \text{ min}} = x \frac{\text{gal}}{\text{min}}$$

At this point we have all our constraints and variables set up, utilizing the solver program in excel we arrive at these results:

$T_2 = 168.15 \text{ } ^\circ\text{C}$

Fluid Conditions	Mixture	C3H8	C7H8	C8H10	C8H10
Molar Composition	1.0000	0.33	0.3	0.15	0.22
Temperature, C	168.1499659				
Pressure, bar	10.0000				
Saturated Vapor Pressure, bar		113.9561846	3.97291476	2.127082623	2.032920806
Saturated Vapor Temperature, C					
Molecular Weight	81.4765	44.0970	92.1410	106.1680	106.1680
K-Values		11.39561846	0.397291476	0.212708262	0.203292081
New K-Values					
Vapor to Feed Molar Ratio [a=V/F]	0.3986				
The Rachford-Rice Equation	0.0000				
Liquid Flow, gpm	700.000				
Liquid Phase					
Molar Composition	1.0000	0.06416	0.39486	0.21860	0.32238
New Molar Composition					
Molar Volume, m ³ /kmol	0.1436	0.20351	0.12830	0.14716	0.14800

Problem 6.A.3 Heat of Condenser

Based on the assumption that there is no pressure change through the condenser, the heat balance is:

$$Q = H_F - \alpha H_V - (1 - \alpha) H_L \quad \text{Where:} \quad (\text{Eq.3.1})$$

$$H_F = \sum_{i=1}^N z_i H_0^{if} + \int_{T_0}^{T_F} \sum z_i C_{pi}^{IG} dT$$

$$H_V = \sum_{i=1}^N y_i H_0^{if} + \int_{T_0}^T \sum y_i C_{pi}^{IG} dT$$

$$H_L = \sum_{i=1}^N x_i H_0^{if} + \int_{T_0}^T \sum x_i C_{pi}^{IG} dT - \sum_{i=1}^N x_i \Delta H_i^{Vap}$$

$$\Delta H_i^{Vap}(T) = \Delta H_i^{Vap}(T_b) \left[\frac{T_c - T}{T_c - T_b} \right]^n$$

Utilizing functions written in 301 Calculator.xlsm the values for the heat of the condenser are easily organized and calculated. Moving from the VLE Calculations to the ENG Calculations tab we have to transfer over a few values. First move over and transpose the molar compositions for the Feed, Vapor and Liquid phases. We then move over the temperatures and pressures for each of these phases. For the Feed and Vapor phase are only temperature dependent and the enthalpy can be calculated using the HIG function, as described by the equations above. The liquid phase enthalpy requires the subtraction of an Hvap term. It should be noted that because the temperature of the liquid phase is above that of the critical temperature of propane we take the Hvap term to be 0 for propane. The sum-product is then taken of these enthalpies and the molar compositions of each phase. We enter these results into Eq.3.1 then multiply by the molar flow rate and convert hours to seconds:

$$Q = 18020.7 \text{ kW}$$

Molar Composition	Units	Feed	Vapor	Liquid
C3H8	mol fraction	0.33	0.73109	0.0642
C7H8		0.3	0.15688	0.3949
C8H10		0.15	0.04650	0.2186
C8H10		0.22	0.06554	0.3224
Conditions and Properties	Units			
Molar Flow	kmol/s			1107.1765
Temperature	C	275.0000	168.1499659	168.1499659
Pressure	bar	10.0000	10.0000	10.0000
Duty of Condenser	kW	18020.70226	Q = H _F - αH _V - (1 - α)H _L	

Problem 6.A.4 Power Requirements of Pump

The energy balance around the pump reduces to:

$$\dot{W} = \frac{\Delta P}{\rho} \cdot \dot{m}$$

$$W_r = \frac{W_i}{\eta}$$

The change in pressure is 10 bar or 1000 kPa, the density is obtained by taking the inverse of the average molar volume which we calculated earlier. We then multiply by the liquid molar flow rate, convert from hours to seconds and then divide by efficiency:

$W_{pump} = 58.88 \text{ kW}$	
L Flow Rate	1107.1765
Delta P	1000.0000
Density	6.9639
Wpump, kW	58.88415039
$\dot{W} = \frac{\Delta P}{\rho} \cdot \dot{m}$	

Problem 6.A.5 Temperature Rise Across Pump

The increase in temperature across the pump can be calculated using numerical methods in excel. Begin by selecting an arbitrary temperature and calculate the liquid enthalpy at that temperature similarly to the method in problem 6.A.3. We can then calculate the work by taking the difference of these liquid enthalpies at the two different temperatures. Multiplying by flow rate, converting and dividing by efficiency we have the work requirement of the pump at this new arbitrary temperature. To find the temperature rise we set the work calculated in this step equal to the work calculated in problem 6.A.4 by varying the temperature:

	T4	T5
	168.1499659	168.7631414
Hliquid	15789.0747	15932.6716
Wpump	58.88415039	58.8842

Extended Secant Method

Make the difference between the two cells equal to the value in the TextBox: 0.0

Changing Cell: 'ENG Calculations'!\$J\$25

Convergence Tolerance: 1.0E-6

Number of Iterations: 30

OK Cancel